Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Pest Management

- **Pest identification:** Documented tomato (*Solanum lycopersicum*) pests in the sources include insect pests, nematodes, rodents, and diseases: tomato fruit borer (*Helicoverpa armigera*), tobacco caterpillar (*Spodoptera litura*), South American tomato pinworm (*Tuta absoluta*), whitefly (*Bemisia tabaci*, order Hemiptera, family Aleyrodidae), aphids, thrips, serpentine leaf miner, red spider mite (*Tetranychus* spp.), leafhoppers, cutworm, root-knot nematode (*Meloidogyne* spp.), lesser bandicoot, palm rat or house rat (*Rattus rattus*), early blight (*Alternaria solani*), late blight (*Phytophthora infestans*), bacterial wilt (*Ralstonia solanacearum*), Fusarium wilt, Septoria leaf spot, bacterial stem and fruit canker, buck eye rot, powdery mildew, damping off, tomato leaf curl disease, tomato bud necrosis virus, and mosaic virus. Documented identifying characteristics and life stages include whitefly pear-shaped, light-yellowish, stalked eggs laid under leaves, oval scale-like greenish-white first-instar nymphs, and tiny white scale-like adults; *S. litura* egg masses covered by brown hairs, velvety black larvae measuring 32–40 mm in length with yellowish-green dorsal stripes, and stout grey-to-dark-brown adult moths; and hyaline, globular eggs hatching into yellowish nymphs and small red adults for red spider mites.

- **Symptoms and damage:** Whitefly nymphs and adults suck sap from leaf lower surfaces, causing chlorotic spots, yellowing, downward curling, leaf drying, and transmission of tomato leaf curl virus. Red spider mites cause reddish-brown bronzing of leaves, silken webbing, leaf drying/withering, and affected flower/fruit formation. *Helicoverpa armigera* and *Spodoptera litura* larvae scrape leaf lamina creating papery white patches in early stages, eat irregular holes in leaves leading to skeletonization and heavy defoliation, and bore irregular holes into fruits. Leaf miners produce live mines on foliage. Late blight produces water-soaked, purple-brown, greasy leaf lesions, grayish-white mycelial rings on leaf undersides, and golden-to-dark brown, firm, sunken fruit spots. Bacterial stem and fruit canker causes white blister-like leaf margin spots, light-coloured stem streaks, bark cracks/cankers, slimy bacterial ooze, and water-soaked fruit spots with white halos. Septoria leaf spot creates small, round-to-irregular spots with grey centers and dark margins on lower foliage leading to blight and defoliation. Root-knot nematodes cause root galls and root rot.

- **Pest biology and favourable conditions:** *Spodoptera litura* females lay approximately 300 eggs in clusters that hatch in 3–5 days; larval development passes through 6 instars lasting 15–30 days, pupation occurs in soil for 7–15 days, and adults live 7–10 days, completing a total life cycle of 32–60 days with 8 generations

per year. Whitefly (*B. tabaci*) eggs hatch in 5–9 days, completing 10–15 generations per year. Red spider mite eggs hatch in 2–6 days, with nymphal stages lasting 3–12 days and adults living 14–21 days. Favourable environmental conditions for whiteflies, tobacco caterpillars, and red spider mites include warm weather and rainy conditions. Septoria leaf spot and bacterial stem/canker are favoured by high humidity, persistent dew, moist weather with intermittent showers, and temperatures of
(25°C
)*and*
(28°C
), *respectively.Bacterialpathogenssurviveprimarilyoninfectedplantdebris*, *seeds*, *andsolanaceou*

- **Pest monitoring and economic thresholds:** Field scouting is conducted at weekly intervals starting after crop establishment across 5 randomly selected spots per field, sampling 5 random plants per spot. Sampling methods require counting nymphs and adults on 5 random leaves per plant for aphids, whiteflies, and mites; 5 terminal leaves per plant (or via tapping) for thrips; live mines on 5 leaves per plant for leaf miners; and total fruit count, damaged fruits, and larval counts per plant for *Helicoverpa* and *Spodoptera*. Nematode soil sampling requires processing 100–300 $cm^3(200 - -300\,g)soilthrough20 -$ $mesh, 200-mesh, and325-meshsieves.Trappingtoolsincludepheromonetraps@4--5traps/acre(or10-$ $-12traps/ha)set1foot(or2--3feet)abovethecanopy(luresreplacedevery2--3weeks); yellowpanwater$ $-5traps/acreset15\,cmabovecanopyforwhiteflies; bluestickytraps@4--5traps/acreforthrips; lighttr$ $-3traps/ha)operatedbetween6\,p.m.and10\,p.m.withsmallnaturalenemyexitoptions; andfruitflytraps@$ $-25traps/ha.DecisionthresholdsrelyonAgro-EcosystemAnalysis(AESA)evaluatingthePest-$ $to-Defender(P:D)ratioacrossparasitoids, predators, andpathogens; explicitnumericaleconomicinj$

- **Preventive and cultural management:** Documented preventive measures include deep summer ploughing to expose pupae/larvae of *Helicoverpa* and *Spodoptera* and reduce nematodes; soil solarization using 45-gauge (
(0.45
) mm) polythene sheets for 3 weeks prior to sowing; raising nurseries in solarized or nematode-free beds; stale seedbed techniques; and crop rotation with non-host cereal crops, cucurbits, or cruciferous vegetables. Ecological engineering strategies involve planting African marigold nursery 15 days prior to tomato as an ovipositional trap crop for *Helicoverpa*, intercropping with beans (to reduce leaf miner), cowpea, onion, maize, coriander, urdbean, or *Ocimum*/basil, and growing 4 rows of maize, sorghum, or bajra as barrier crops around the field. Sanitation and cultural care include field sanitation, destroying crop debris, using raised beds (15 cm height) or pro-trays, avoiding excessive watering/logging, maintaining plant spacing of approximately 3 feet, applying black LDPE mulch (30 micron thickness buried 10 cm deep), staking plants, clipping lower leaves touching the ground (up to 30 cm), and incorporating FYM @ 8–10 t/acre, vermicompost @ 5 t/acre, or neem cake @ 100 kg/acre.

- **Biological, mechanical, and botanical control:** Documented biological control agents include inundative release of the egg parasitoid *Trichogramma pretiosum* @ 20,000–40,000/acre/week (4–5 releases from flowering), or *Trichogramma achaea / Trichogrammatoidea bactrae* @ 50,000/ha (6 releases at 7-day intervals against pinworm); whitefly parasitoids (*Encarsia formosa*, *Eretmocerus* spp., *Chrysocharis pentheus*); predators such as green lacewing larvae @ 4000/acre, ladybird beetles (*Coccinella septempunctata*,

Menochilus sexmaculata), big-eyed bugs (*Geocoris* sp.), mirid bugs (*Dicyphus hesperus*), and encouraging predatory birds by erecting bird perches @ 20/acre. Biopesticides include seed treatment with *Trichoderma viride* 1% WP @ 9 g/kg seed or root-zone application of 2.5 kg *T. viride* 1% WP in 150 kg FYM; foliar spraying of *Beauveria bassiana*, *Lecanicillium lecanii*, or *Metarhizium anisopliae* @
$(1 \times 10^8$
) cfu/ml; $Bacillus\ thuringiensisvar.kurstaki@500\,g/ha; andentomopathogenicnematodes(Steinernema\ f$
$(m^2$
$)orperacre.Mechanicalcontrolinvolveshandpickinganddestroyingeggmasses, larvae, andinfectedfrui$
$mixedwater, settingbonfiresat7--8\,p.m., smokingrodentburrowsfor2--3minutes, erectingowlperch$
$-6/acre, andinstallingmulti-catchwondertraps@8--10traps/acre.BotanicalmanagementincludesN$

- **Chemical pest management:** Chemical treatments for seedling root dip include dipping roots for 15 minutes in imidacloprid 17.8% SL @ 60–70 ml in 200 l water/acre prior to transplanting. Insecticides and acaricides include cyantraniliprole 10.26% OD @ 360 ml in 200 l water/acre; dimethoate 30% EC @ 264–396 ml in 200–400 l water/acre; thiamethoxam 25% WG @ 80 g in 200 l water/acre; malathion 50% EC @ 600 ml in 200–400 l water/acre; carbofuran 3% CG @ 16,000 g/acre; phorate 10% CG @ 6,000 g/acre; fenazaquin 10% EC @ 500 ml in 200 l water/acre or spiromesifen 22.9% SC @ 250 ml in 200 l water/acre (against red spider mites); indoxacarb 14.5% SC @ 160–200 ml in 120–240 l water/acre; flubendiamide 20% WG @ 40 g or flubendiamide 39.35% M/M SC @ 40 ml in 150–200 l water/acre; carbaryl 50% WP @ 800 g in 200–400 l water/acre; chlorantraniliprole 18.5% SC @ 60 ml in 200 l water/acre; lambda-cyhalothrin 4.9% CS @ 120 ml in 200 l water/acre or 5% EC @ 120 ml in 160–200 l water/acre; methomyl 40% SP @ 300–450 ml in 200–400 l water/acre; novaluron 10% EC @ 300 ml in 200–400 l water/acre; novaluron 5.25% + indoxacarb 4.5% SC @ 330–350 ml in 200 l water/acre; quinalphos 20% AF @ 600–700 ml or 25% EC @ 400 ml in 200–400 l water/acre; and dazomet technical @ 12–16 g/acre for nursery soil nematodes. Fungicides include mancozeb 35% SC @ 200 ml in 200 l water/acre or mancozeb 75% WP @ 600–800 g in 300 l water/acre; zineb 75% WP @ 600–800 g in 300–400 l water/acre; famoxadone 16.6% + cymoxanil 22.1% SC @ 200 ml in 200 l water/acre; cymoxanil 8% + mancozeb 64% WP @ 600 g in 200–300 l water/acre; azoxystrobin 23% SC @ 200 ml in 200 l water/acre; captan 50% WP @ 1000 g in 300–400 l water/acre or captan 75% WP @ 666.8 g in 400 l water/acre; copper oxychloride 50% WP @ 1000 g in 300–400 l water/acre; cyazofamid 34.5% SC @ 80 ml in 200 l water/acre; metiram 75% WG @ 1000 g in 200–300 l water/acre; and propineb 70% WP @ 120 g in 40 l water/acre. Rodent control utilizes zinc phosphide poison bait (96 parts broken tomato + 2 parts edible oil + 2 parts zinc phosphide) @ 10 g/burrow following pre-baiting with 20 g plain bait/burrow.

- **Integrated Pest Management (IPM):** IPM strategies combine cultural, mechanical, biological, and chemical methods sequentially across crop growth stages based on weekly morning AESA field observations before 9 a.m. Interventions prioritize pre-sowing solarization, raised nursery beds, biopesticide seed treatment, trap crop establishment, and barrier crops; vegetative stage parasitoid releases, predator conservation, sticky/light traps, and botanical sprays; and reproductive stage pheromone traps and targeted biointensive or chemical applications. Chemical sprays are restricted to prevent natural enemy destruction and avoided when 1–2 larval parasitoids are observed or within 7 days of parasitoid release.

- **Application safety and resistance management:** Documented safety guidelines mandate choosing short-persistent pesticides to prevent residues in soil and produce, with complete prohibition of chemical applications during the 7 days preceding harvest. Spraying must thoroughly cover leaf undersides, especially for mites, whiteflies, and *Spodoptera*. Midday spraying should be avoided when insects are inactive; nocturnal pests (*Helicoverpa*, *Spodoptera*) and nuclear polyhedrosis viruses (HaNPV/SlNPV) should be treated in the evening after 5 p.m. To manage pesticide resistance and cross-resistance (defined by IRAC as a heritable change in pest population sensitivity causing repeated control failure), growers must rotate pesticide modes of action and avoid applying long-persistent chemical pesticides on trap crops to maintain pest and natural enemy attraction. Personal protective equipment details are Not specified in the provided sources.
- **Management efficacy and economic impact:** Not specified in the provided sources.
- **Post-treatment monitoring and recordkeeping:** Post-treatment evaluation is conducted through weekly follow-up field scouting using AESA parameters to record pest reinfestation, defender populations, and percentage fruit damage. Pheromone trap surveillance requires logging weekly moth counts of *H. armigera* and *S. litura* year-round, with trapped moths destroyed after each recording. Standardized post-treatment evaluation forms or formal application recordkeeping templates are Not specified in the provided sources.
- **Conflicting information and limitations:** Documented variations exist across source manuals regarding recommended trap densities and bio-agent release rates, including pheromone trap densities specified as 4–5 traps/acre in AESA manuals versus 10–12 traps/ha in Good Agricultural Practices (GAP) guidelines, light trap recommendations specified as 1 trap/acre versus 2–3 traps/ha, and *Trichogramma pretiosum* inundative release rates reported as 20,000/acre/week, 40,000/acre, or 50,000/ha. Documented limitations include the skill requirement for AESA observations (requiring trained farmers/facilitators) and the legal exemption from registration for biopesticides produced by farmers for on-farm consumption.